

MATH3881/5881: Statistical Machine Learning Theory

Week 3: Theory Tutorial Solutions: Modified Dropout as Ensemble Averaging

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Worked solution to the Week 3 Theory Tutorial. Numbering matches the problem sheet.

Problem: Modified dropout is unbiased gradient descent in expectation

Part (a).

Step 1. By linearity of expectation, the i -th component of $\tilde{g}$ has

$$\mathbb{E}[\tilde{g}_i] = \mathbb{E}\left[\frac{1}{q} m_i \cdot (\nabla C(\theta))_i\right] = \frac{1}{q} \mathbb{E}[m_i] \cdot (\nabla C(\theta))_i,$$

where the gradient component $(\nabla C(\theta))_i$ is treated as a constant because the mask is independent of θ .

Step 2. Since $m_i \sim \text{Bern}(q)$, we have $\mathbb{E}[m_i] = q$, so

$$\mathbb{E}[\tilde{g}_i] = \frac{1}{q} \cdot q \cdot (\nabla C(\theta))_i = (\nabla C(\theta))_i.$$

Step 3. The same calculation holds componentwise, so $\mathbb{E}[\tilde{g}] = \nabla C(\theta)$.

Step 4. Therefore one modified-dropout step is, in expectation,

$$\mathbb{E}[\theta - \alpha \tilde{g}] = \theta - \alpha \nabla C(\theta),$$

which is exactly one standard gradient-descent step (Week 1) with step size α . The modified scheme is unbiased.

Part (b).

Step 1. Repeating the Step 1 calculation of part (a) but without the $1/q$ factor,

$$\mathbb{E}[\tilde{g}'_i] = \mathbb{E}[m_i] \cdot (\nabla C(\theta))_i = q \cdot (\nabla C(\theta))_i,$$

so $\mathbb{E}[\tilde{g}'] = q \nabla C(\theta)$.

Step 2. One raw-mask step is therefore, in expectation, a standard gradient-descent step (Week 1) with the *rescaled* step size αq , not α . The $1/q$ factor (used by inverted dropout, including `nn.Dropout`) is exactly the correction that cancels this rescaling, so the user can set the learning rate α without having to compensate for the dropout rate.

Part (c).

Step 1. For the linear model $f(\theta; x) = \theta^\top x$, the right-hand side is

$$\mathbb{E}_m \left[f\left(\frac{1}{q} m \odot \theta; x\right) \right] = \mathbb{E}_m \left[\left(\frac{1}{q} m \odot \theta\right)^\top x \right] = \mathbb{E}_m \left[\sum_{i=1}^d \frac{m_i}{q} \theta_i x_i \right].$$

Step 2. By linearity of expectation and $\mathbb{E}[m_i] = q$,

$$\mathbb{E}_m \left[\sum_{i=1}^d \frac{m_i}{q} \theta_i x_i \right] = \sum_{i=1}^d \frac{\mathbb{E}[m_i]}{q} \theta_i x_i = \sum_{i=1}^d \theta_i x_i = \theta^\top x = f(\theta; x),$$

which gives the claimed equality. In words: the standard test-time prediction of a linear model trained with this modified dropout is *exactly* the average prediction over all 2^d thinned subnetworks. This is the linear case of the “dropout is an ensemble of thinned subnetworks” claim, and the empirical content of Problem 2 of the Practice Tutorial is that the same approximate equality continues to hold for the nonlinear feedforward network with standard dropout, trained on MNIST.